

软件工程概论

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课程成绩构成：

(1) 平时成绩50%：

- 10%：考勤和课堂讨论；
- 20%：个人完成软件工程实训；
- 20%：个人随堂测验（软件工程综合知识）；

(2) 期末成绩50%：

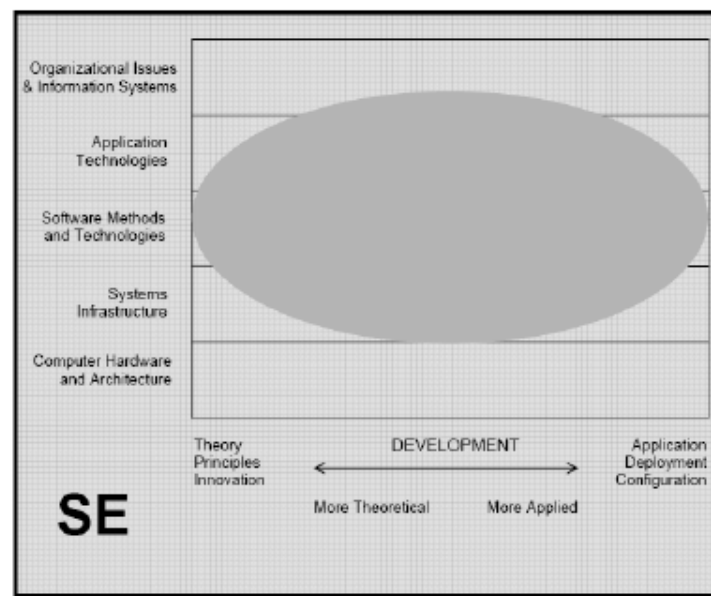
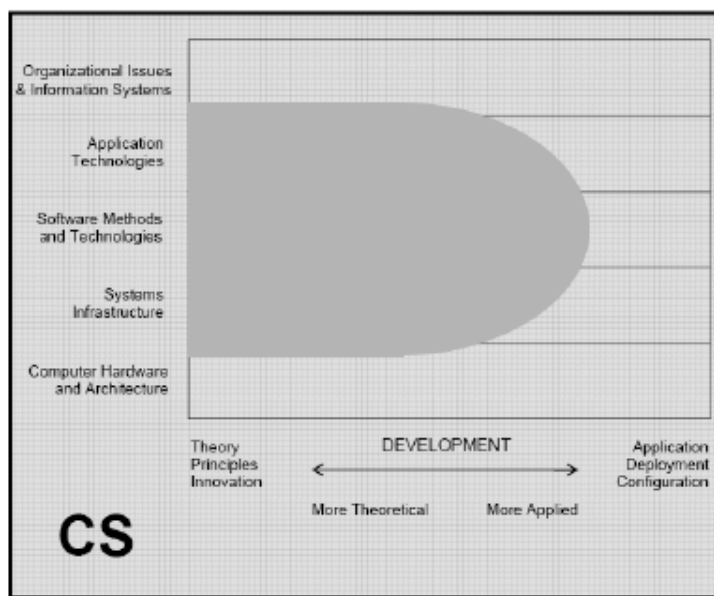
- 50%：个人完成一个软件工程项目；
 - 项目完成情况35%、报告质量15%。

- 软件工程项目：
 - 功能需求+非功能需求;
 - 必须有数据存储：关系/非关系/图/混合数据库;
 - 必须有UI，建议用成熟框架;
 - 建议考虑AI的融合;
 - 项目包含完整生命周期;

软件工程学科

内容取自清华大学 刘强

- 争议一：软件工程与计算机科学在问题空间上的界定
 - 二者是存在一定交集的两个独立部分？
 - 软件工程是计算机科学的一个子集？
- **IEEE-CS/ACM Computing Curricula 2005**



- 争议二：软件工程与计算机科学在教育范畴上的界定
 - 软件工程是工程领域的一种门类？
 - 软件工程是计算机科学的一部分？
- 软件工程专家观点
 - **Fred Brooks:** *A scientist builds in order to learn; an engineer learns in order to build.*
 - **David Parnas :** *Software engineering programs are not computer science programs.*
 - **Manfred Broy:** *Engineering disciplines must be based on scientific practices and theory to justify their approaches and to give scientific evidence for why and where their methods work properly.*

软件工程实践

The Essence of Practice

- Suggests:
 1. *Understand the problem* (communication and analysis).
 2. *Plan a solution* (modeling and software design).
 3. *Carry out the plan* (code generation).
 4. *Examine the result for accuracy* (testing and quality assurance).

Understand the Problem (requirements analysis)

- *Who has a stake in the solution to the problem?* That is, who are the stakeholders?
- *What are the unknowns?* What data, functions, and features are required to properly solve the problem?
- *Can the problem be compartmentalized (分解)?* Is it possible to represent smaller problems that may be easier to understand?
- *Can the problem be represented graphically?* Can an analysis model be created? (yes)

Plan the Solution (case study and design)

- *Have you seen similar problems before?* Are there patterns that are recognizable in a potential solution? Is there existing software that implements the data, functions, and features that are required?
- *Has a similar problem been solved?* If so, are elements of the solution reusable?
- *Can subproblems be defined?* If so, are solutions readily apparent for the subproblems?
- *Can you represent a solution in a manner that leads to effective implementation?* Can a design model be created?

Carry Out the Plan (coding)

- *Does the solution conform to the plan?* Is source code traceable to the design model?
- *Is each component part of the solution provably correct?* Has the design and code been reviewed, or better, have correctness proofs been applied to algorithm?

Examine the Result (test)

- *Is it possible to test each component part of the solution?* Has a reasonable testing strategy been implemented?
- *Does the solution produce results that conform to the data, functions, and features that are required?* Has the software been validated against all stakeholder requirements?

谢谢!